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Elevated Pulmonary Artery Wedge Pressure in Group 1 Pulmonary Hypertension

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BACKGROUND: With pulmonary hypertension (PH), a pulmonary artery wedge pressure (PAWP)>15 mm Hg is used to diagnose left heart dysfunction, but some patients with adjudicated group 1 PH demonstrate PAWP>15 mm Hg. The primary objective of the study was to evaluate group 1 PH with high PAWP>15 mm Hg.

METHODS: Patients with adjudicated group 1 PH from PVDOMICS between 2016 and 2019 were separated into high PAWP(>15 mm Hg) or normal PAWP and compared with adjudicated combined pre- and postcapillary (Cpc) PH related to heart failure with preserved ejection fraction (HFpEF). Participants underwent dynamic right heart catheterization and metabolomics. Findings were validated in 3 independent cohorts with adjudicated group 1 PH (validation cohorts 1 and 2 with exercise right heart catheterization and validation cohort 3 with resting right heart catheterization).

RESULTS: Of 325 patients with group 1 PH (73% women, mean age 53.0±14.3 years), 15% (n=48) had high PAWP. Group 1 PH+high PAWP demonstrated greater obesity, left ventricular hypertrophy (P<0.0002), left ventricular strain impairment (P<0.0001), and decreased left ventricular compliance (P<0.0001) compared with group 1 PH+normal PAWP, with changes comparable to Cpc-PH HFpEF (n=75). Compared with Cpc-PH HFpEF, left atrial function was better in group 1 PH+high PAWP with lower PAWP V wave, higher left atrial compliance, and left atrial ejection fraction (P<0.0001 for all). Metabolomics demonstrated little difference between group 1 PH with high versus normal PAWP but large differences between group 1 PH+high PAWP versus Cpc-PH HFpEF (100 metabolites altered at false discovery rate P<0.05). Elevated PAWP was also observed in 18% of group 1 PH in the validation cohort 1 (n=402), with exercise PAWP response intermediately abnormal in group 1 PH+high PAWP relative to Cpc-PH HFpEF and group 1 PH+normal PAWP (interaction P<0.0001). Elevated PAWP was similarly observed in 22% and 19% of group 1 PH in validation cohorts 2 (n=55) and 3 (n=787), respectively.

CONCLUSIONS: Approximately 1 in 5 adjudicated patients with group 1 PH has elevation in resting PAWP despite severe pulmonary vascular dysfunction and metabolomics consistent with traditionally defined group 1 PH. Despite resting PAWP elevation, these patients with group 1 PH were metabolomically and biologically distinct from Cpc PH HFpEF, with better left atrial function and diastolic reserve during exercise. These data emphasize the limitations of using resting PAWP alone to separate group 1 PH from HFpEF and call for development of more integrated clinical diagnostic criteria.

Key Words: hemodynamics ■ PAWP ■ pulmonary hypertension

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Patients with group 1 pulmonary hypertension (PH) have a primary pulmonary vasculopathy that leads to elevation in pulmonary vascular resistance (PVR). Since left heart disease and heart failure with preserved ejection fraction (HFpEF) can also cause elevation of PVR, guidelines have recommended a resting pulmonary artery wedge pressure (PAWP) threshold of 15 mm Hg to diagnose left heart dysfunction based on reported PAWP values from healthy controls.[1] The cause of PH and associated PVR elevation

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Clinical Perspective

What Is New?

- Approximately 1 in 5 patients with clinically adjudicated group 1 pulmonary hypertension (PH) has an elevated resting pulmonary artery wedge pressure (PAWP)>15 mm Hg, which is commonly used to hemodynamically diagnose heart failure with preserved ejection fraction and exclude group 1 PH.
- Despite having a PAWP>15 mm Hg, clinical features, metabolomics, and pulmonary vascular dysfunction in group 1 PH with high PAWP were most consistent with a true diagnosis of group 1 PH as opposed to heart failure with preserved ejection fraction.

What Are the Clinical Implications?

- A resting PAWP>15 mm Hg alone should not be used to exclude a diagnosis of group 1 PH, with the diagnosis instead requiring probabilistic evaluation of all available clinical information (including hemodynamics) in adjudication.
- By virtue of existing trial criteria, patients with group 1 PH and high PAWP have been excluded from all prior trials, creating uncertainty in optimal management.
- These data suggest an unmet need for more probabilistic diagnostic criteria for group 1 PH on the basis of consideration of all clinical features as opposed to strict dichotomization by resting hemodynamics alone.

Nonstandard Abbreviations and Acronyms

BMI	body mass index
Cpc	combined pre- and postcapillary
HFpEF	heart failure with preserved ejection fraction
LA	left atrial
LV	left ventricular
LVTMP	left ventricular transmural pressure
mPA	mean pulmonary artery
PAWP	pulmonary artery wedge pressure
PH	pulmonary hypertension
PVR	pulmonary vascular resistance
RA	right atrial

is therefore currently dichotomized by resting PAWP, where patients with a PAWP≤15 mm Hg are considered to have group 1 PH (after excluding lung-related [group 3] or thromboembolism-related [group 4] PVR elevation), whereas those with PAWP>15 mm Hg are diagnosed with left heart–related PVR elevation; ie, combined pre- and postcapillary (Cpc) PH HFpEF.[2,3]

As with any binary dichotomization process, use of a single resting PAWP cutoff may introduce disease misclassification.[4–6] A subset of patients is diagnosed and treated as group 1 PH despite PAWP>15 mm Hg,[7–9] but such patients are routinely excluded from clinical trials that mandate a resting PAWP≤15 mm Hg.[10] We hypothesized that patients with group 1 PH with PAWP>15 mm Hg would have precapillary pulmonary vascular pathology more similar to traditional group 1 PH and distinct from Cpc PH HFpEF. To test our hypothesis, we used the multicenter PVDOMICS dataset, where all individuals underwent hemodynamic provocation, manual readjudication of all recorded pressures at end expiration, unbiased metabolomics evaluation, and integrated PH specialist adjudication of the diagnosis of group 1 PH with normal PAWP, group 1 PH despite high PAWP, or Cpc PH HFpEF. We also independently tested the prevalence and pathophysiology of resting PAWP elevation in group 1 PH among 2 independent validation cohorts undergoing exercise right heart catheterization (RHC) and a third validation cohort undergoing resting RHC, with all patients undergoing probabilistic adjudication of group 1 PH or Cpc PH HFpEF by PH specialists.

METHODS

Primary PVDOMICS Cohort Description

The PVDOMICS clinical research network is a National Heart, Lung, and Blood Institute–funded, prospective, longitudinal cohort study (NCT02980887) that enrolled participants from November 30, 2016 to October 18, 2019 at 7 centers across the United States.[11] The protocol was approved at each institution by the local institutional review board, and informed consent was obtained from all participants.

Case Definitions

All participants in PVDOMICS underwent formal adjudication of their "primary" World Symposium on Pulmonary Hypertension group using integrated probabilistic assessment of all available diagnostic information. Initial group assignments were done by the recruiting investigators who were practicing PH physicians at each site. To avoid tautological bias inherent to classification by resting PAWP thresholds alone, patients were allowed to be categorized as group 1 PH despite a PAWP>15 mm Hg. However, additional adjudication of this subset of patients was performed by an independent PVDOMICS adjudication committee.

The final PVDOMICS-adjudicated PH diagnosis was based on the clinical, imaging, and hemodynamic profiles. This included resting and provoked hemodynamics, severity of PAWP and PAWP V wave elevation relative to mean pulmonary artery (mPA) pressure and right atrial (RA) pressure, severity of pulmonary vascular dysfunction, prior history and clinical response to PH vasodilator therapy, and comprehensive cardiac imaging. The full consensus criteria used are detailed in the Supplemental Material and summarized in Table 1.

Group 1 PH was therefore defined as mPA>20 mm Hg with PVDOMICS-adjudicated "primary" group 1 PH. For the main

Table 1. Factors Considered in Probabilistic Adjudication of Group 1 Pulmonary Hypertension or Combined Pre- and Postcapillary Pulmonary Hypertension From Heart Failure With Preserved Ejection Fraction

	Supportive of group 1 PH	Supportive of Cpc PH HFpEF
Clinical diagnoses or risk factors	Presence of clear risk factors: Known genetic mutation or family history of group 1 PH Known drug/toxin exposure such as amphetamines Known congenital heart disease that predisposes to precapillary PH, such as left to right shunt lesion Other known risk factors, such as connective tissue disease and portal hypertension	Presence and severity of clinical risk factors for HFpEF: Older age Atrial fibrillation Obesity Diabetes Coronary disease Hypertension
Severity of PA pressure elevation relative to PAWP	More severe PH Higher transpulmonary gradient and PVR, even if PAWP mildly elevated at end expiration	Less severe PH Higher PAWP with lower transpulmonary gradient and PVR
Qualitative analysis of PAWP V wave and RA pressure	Flat PAWP at end expiration without prominent V wave, with prominent drop in PAWP with inspiration RA pressure elevated and exceeding or equalized with PAWP at end expiration	Large V wave in PAWP at end expiration PAWP exceeding RA pressure
Response of PAWP and PA pressure to provocation with nitric oxide, fluid, or exercise	Low or normal PAWP regardless of provocation Lack of increase in v wave during provocation RA pressure elevated, exceeding or equal to PAWP Disproportionate rise in PA pressure with provocation relative to PAWP	Very high PAWP with provocation with increase in PAWP V wave relative to baseline Relatively proportional rise in PA and PAWP with exercise
Prior pulmonary vasodilator use before study RHC	History of pulmonary vasodilator use (especially parenteral prostaglandins) with objective clinical improvement.	No prior pulmonary vasodilator use or monotherapy (especially PDE5 inhibitor) without clinical improvement
Imaging studies	Disproportionate right heart remodeling Small, underfilled, D-shaped left ventricle Normal sized left atrium	Left atrial enlargement Functional mitral regurgitation

Cpc indicates combined pre- and postcapillary; HFpEF, heart failure with preserved ejection fraction; PA, pulmonary artery; PAWP, pulmonary artery wedge pressure; PDE5, phosphodiesterase type 5; PH, pulmonary hypertension; PVR, pulmonary vascular resistance; and RHC, right heart catheterization.

analysis, patients with group 1 PH were dichotomized based on resting PAWP, grouping those with a normal PAWP≤15 mm Hg (group 1 PH+normal PAWP) and those with an abnormal PAWP>15 mm Hg (group 1 PH+high PAWP).

As an additional reference population, we also included patients with Cpc PH due to HFpEF. We focused on the HFpEF population rather than group 2 PH in general, as these patients are most at risk for misdiagnosis/overlap with group 1 PH.[8] Included patients with Cpc PH HFpEF met resting hemodynamic criteria (mPA>20 mm Hg, PAWP>15 mm Hg, and PVR>2 Wood units) and were independently adjudicated by PVDOMICS investigators as group 2 PH related to HFpEF. As an agnostic, adjudication-independent estimate of the likelihood of HFpEF, we also quantified pretest probability for HFpEF using the validated HFpEF-ABA (age, body mass index, and atrial fibrillation) algorithm based only on clinical characteristics.[12] Healthy asymptomatic controls recruited to PVDOMICS formed another reference population for normative values of imaging measures (Figure S1).

Clinical and Hemodynamic Evaluation

Participants underwent detailed baseline evaluation, including imaging, functional and hemodynamic characterization, as described previously (see details in the Supplemental Material). RHC was performed at rest in the supine position with manual readjudication of pressure measurements at end expiration, as described previously.[13] Patients then underwent repeat hemodynamic assessment during (1) 100% oxygen challenge for 5 minutes and (2) inhaled nitric oxide challenge at 40 ppm for 5 minutes added to 100% oxygen with assessment of PAWP, mPA, and cardiac output by thermodilution in both phases. The PAWP V wave was manually measured at end expiration as the highest point on the PAWP tracing timed around the ECG T wave. Hemodynamic stress for left heart disease was performed on the basis of center availability either through (1) fluid challenge with 500 mL of 0.9% saline over 10 minutes or (2) invasive exercise hemodynamics to peak exhaustion in the supine or upright position based on clinical practice patterns at recruiting centers.

Assessment of Left Heart Diastolic Function

The contribution of myocardial stiffness of the left ventricle to PAWP elevation was estimated using cardiac magnetic resonance imaging and RHC derived stiffness constant β (increases with greater left ventricular [LV] stiffness), and the estimated LV end-diastolic volume at a common PAWP of 30 mm Hg (decreases with greater LV stiffness). Left atrial (LA) operating compliance and other LA function indices were calculated as described previously (see details in the Supplemental Material).[14,15]

Metabolomics measurement

All PVDOMICS participants underwent fasting venous plasma collection for untargeted metabolomics as previously described.[16](details in Supplemental Material)

Validation Cohorts 1 to 3

Validation cohort 1 included consecutive patients undergoing supine exercise RHC at the Mayo Clinic between 2006 and 2024. PH diagnosis was adjudicated independently by Y.N.V.R. and W.R.M. (for details, see the Supplemental Material). Catheterization included simultaneous gas exchange measures, high-fidelity micromanometers, and exercise hemodynamics, as described previously.[17,18] All tracings were manually remeasured at end expiration by a single cardiologist with experience in hemodynamic interpretation (Y.N.V.R.) while blinded to clinical information. From simultaneous RA and PAWP measures, the contribution of pericardial restraint to PAWP elevation was estimated by increasing RA/PAWP ratio and a decrease in LV transmural pressure (LVTMP=PAWP−RA) (see additional details in the Supplemental Material).[19–23]

Similar to the PVDOMICS cohort, the adjudication of group 1 PH was made probabilistically using all available clinical information and detailed evaluation of hemodynamic tracings at rest and exercise. Adjudication of group 1 PH with resting PAWP>15 mm Hg was allowed if the final diagnosis was most consistent with group 1 PH on probabilistic adjudication. As in the other cohorts, prior use of vasodilators alone was insufficient to qualify for a diagnosis of group 1 PH, and the ultimate diagnosis considered all available clinical and hemodynamic factors. As a comparator group, we also included patients with definite Cpc PH HFpEF at rest (mPA>20 mm Hg, PVR>2 Wood units, resting PAWP>15 mm Hg), along with persistent elevation in PAWP during exercise ≥25 mm Hg.

Validation cohort 2 included patients from the Mayo Clinic with adjudicated group 1 PH who underwent exercise RHC between 2024 and 2025 with the exact same criteria as above (adjudication by Y.N.V.R. and W.R.M.).

Validation cohort 3 included patients between 2000 and 2020 from the Mayo Clinic who underwent resting RHC with a prior or new prescription for PH medication following RHC. From this cohort of PH therapy–treated patients, the historical PH provider–adjudicated diagnosis was abstracted by detailed chart review by an investigator (R.V.), followed by independent readjudication by Y.N.V.R. of all cases and hemodynamic tracings as most consistent with either (1) group 1 PH or (2) Cpc PH HFpEF despite use of PH therapy.

Statistical Analysis

The data that support the findings of this study are available from the corresponding author upon reasonable request. Differences across the groups were compared using ANOVA, Kruskal-Wallis test, and chi-square test as appropriate. Individual group comparisons were performed by Tukey and Steel-Dwass test as appropriate in the setting of multiple comparisons. Missing values were not imputed for descriptive analyses. To account for the correlated nature of repeated hemodynamic measures, we used a mixed-model approach with a compound symmetry covariance structure, using participant ID as a random effect, with group, phase, and their interaction as fixed effects. Cpc PH HFpEF and group 1 PH patients stratified by resting PAWP were followed for all-cause mortality using the log-rank test from time of study enrollment. Cox proportional hazard models were used to determine the hazard ratio for death between groups.

Details of metabolomics data preparation for analyses are included in the Supplemental Material. Following metabolite data normalization, we performed univariate 2-sample *t* tests to identify significantly changed metabolites, comparing group 1 PH+high PAWP versus group 1 PH+normal PAWP, and separately compared the metabolome between group 1 PH+high PAWP versus Cpc PH HFpEF. The Benjamini-Hochberg method was used to correct the false discovery rate for the 954 metabolites tested with an adjusted $P<0.05$ considered statistically significant. Given the importance of obesity to metabolome changes,[24–26] we also present metabolite differences adjusted for baseline body mass index (BMI) using general linear models with covariate adjustment (Limma package). Pathway enrichment analysis was performed using metabolite set enrichment overrepresentation analysis. We used the list of all measured metabolites with KEGG (Kyoto Encyclopedia of Genes and Genomes) IDs as a reference metabolome for enrichment tests. We selected the RaMP-DB (Relational Database of Metabolomics Pathways)–based pathway database for enrichment analysis, integrating pathways from KEGG, HMDB (Human Metabolome Database), Reactome, and Wiki Pathways.[27] We evaluated all significant upregulated or downregulated metabolites separately for pathway analysis. The current report adhered to STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) reporting guidelines. Statistical analyses were performed using MetaboAnalyst 6.0 software and JMP version 17.1.0 (JMP Statistical Discovery LLC).

RESULTS

Baseline Characteristics of the Primary PVDOMICS Cohort

Of 325 patients with adjudicated group 1 PH (mean age 53.0±14.3, women 73% [n=236]), 85% (n=277) had normal resting PAWP, and 15% (n=48) had high resting PAWP (Figure S1). Compared with both group 1 PH subgroups, patients with Cpc PH HFpEF (n=75) were older (mean age 68.9±11.4, 64% women [n=23]) with greater hypertension and diabetes but also had the highest natriuretic peptide levels and diuretic usage, worst renal function, greatest inflammation (C-reactive protein), and lowest hemoglobin levels (Table 2). There was a progressive increase in atrial fibrillation burden and obesity from group 1 PH+normal PAWP to group 1 PH+high PAWP to Cpc PH HFpEF, with an associated increase in pretest probability for HFpEF on the basis of the HFpEF-ABA algorithm (Figure S2). Most patients with group 1 PH had known treated PH before enrollment with a high rate of baseline pulmonary vasodilator use (96% [n=46]) in group 1 PH+high PAWP and longer duration of known PH before enrollment.

Cardiac Structure and Function

Patients with Cpc PH HFpEF had the largest LV and LA volumes, greater LV diastolic dysfunction (lower lateral e′ and higher E/e′), and greater LA dysfunction (lower LA EF and expansion index) compared with both group 1 PH subgroups (Table 3; Table S1; Figure S3). Patients with Cpc PH HFpEF also had larger RA volumes and greater impairment in RA EF, consistent with more biatrial myopathy in this group (Table S1).

Table 2. Baseline Characteristics Across Group 1 Pulmonary Hypertension by Resting Pulmonary Artery Wedge Pressure Compared With Combined Pre- and Postcapillary Pulmonary Hypertension Heart Failure With Preserved Ejection Fraction in the PVDOMICS Cohort

	Gp1 PH+PAWP≤15 mm Hg (n=277)	Gp1 PH+PAWP>15 mm Hg (n=48)	Cpc PH HFpEF (n=75)	*P* value
Age, y (n=407)	53.2±14.6	52.2±12.2	68.9±11.4*	<0.0001
BMI mass index, kg/m² (n=407)	28.8±7.2*	32.2±8.7	34.6±8.0	<0.0001
Obesity (BMI ≥ 30 kg/m²)	36	54	73	<0.0001
Persistent atrial fibrillation, %	1	10	29	<0.0001
Women, %	73	69	64	0.28
Race and ethnicity, White/Black/Other‡, %	78/11/11	81/15/4	88/8/4	0.12
Waist circumference, cm (n=345)	95.2±20.7*	108.5±22.3	113.2±22.0	<0.0001
Waist/height ratio	0.58±0.12*	0.66±0.136	0.68±0.13	<0.0001
Waist/height ≥0.5	75	93	93	0.0002
Body fat, % (n=276)	35.3 [27.1–43.0]	38.1 [29.7–46.8]	40.0 [31.8–47.6]†	0.05
Connective tissue disease, %	29	15	7	<0.0001
Coronary disease, %	5	2	16	0.005
Hypertension, %	36	31	72	<0.0001
Diabetes, %	17	21	43	<0.0001
HFpEF-ABA probability	38.7 [19.0–38.7]*	51.8 [27.3–75.9]*	79.8 [64.5–95.8]*	<0.0001
Never/prior/current smoker, %	61/25/4	42/56/2	51/48/1	0.03
Obstructive sleep apnea, %	17	17	32	0.01
Apnea hypopnea index, events/h (n=221)	5.0 [1.2–11.9]	8.0 [1.2–24.2]	9.4 [4.2–27.2]†	0.006
Lab testing				
Hemoglobin, g/dL (n=387)	13.7±1.9	14.1±2.8	12.6±1.9*	<0.0001
GFR, mL/min per 1.73 m² (n=367)	81.0±23.8	78.7±25.9	59.0±20.1*	<0.0001
NT-proBNP, pg/mL (n=390)	242 [93–957]	333 [120–1072]	772 [339–1718]*	<0.0001
Baseline medications/PH risk				
Diuretics, %	62	77	85	0.0001
Phosphodiesterase-5 inhibitor, %	64	75	19	<0.0001
Prevalent PH medication before enrollment, %	79	96	21	<0.0001
PH at enrollment, y	2.7 [0.6–7.6]*	5.3 [2.1–9.4]*	0.2 [0.0–2.4]*	<0.0001
Reveal Lite 2 score	5.4±2.8	5.8±2.5	7.6±2.2*	<0.0001

Values represent mean (SD) or median [25th–75th percentile]. *P* values were computed using overall group ANOVA or Kruskal-Wallis test with post hoc comparisons using Tukey or Steel-Dwass, respectively. ABA indicates age, body mass index, atrial fibrillation; BMI, body mass index; Cpc, combined pre- and postcapillary; GFR, glomerular filtration rate; Gp1, group 1; HFpEF, heart failure with preserved ejection fraction; NT-proBNP, N-terminal pro-B-type natriuretic peptide; PAWP, pulmonary artery wedge pressure; PH, pulmonary hypertension; and PVDOMICS, redefining pulmonary hypertension through pulmonary vascular disease phenomics.

*P<0.05 vs all groups.

†P<0.05 vs Gp1 PH+PAWP≤15 mm Hg.

‡Other indicates non-White or non-Black race and ethnicity.

Group 1 PH+high PAWP had the largest right ventricular size and the poorest RV-PA coupling compared with other groups, while pulmonary artery enlargement was highest in both group 1 PH subgroups. Group 1 PH+normal PAWP had the lowest LV mass and highest LV longitudinal strain (Table 3).

Cardiac Remodeling in Group 1 PH+High PAWP Compared With Healthy Controls

To contextualize the cardiac remodeling observed in group 1 PH+high PAWP, we next contrasted this group with asymptomatic healthy controls (n=96) recruited to PVDOMICS. Group 1 PH+high PAWP was associated with greater obesity, modestly higher hypertension, worse renal function, and higher pretest probability of HFpEF (Table S2). As expected, right heart remodeling and function indices by echocardiogram and magnetic resonance imaging were all abnormal relative to healthy controls. Notably, measures of left heart performance and remodeling were also abnormal in group 1 PH+high PAWP compared with healthy controls, including LV mass, LV longitudinal strain, LA size, LA EF, LA expansion index, e′ velocity, and E/e′ (Figure S3; Table S3).

Table 3. Echocardiography in Group 1 Pulmonary Hypertension by resting Pulmonary Artery Wedge Pressure and Combined Pre- and Postcapillary Pulmonary Hypertension Heart Failure With Preserved Ejection Fraction in the PVDOMICS Cohort

	Gp1 PH+PAWP≤15 mm Hg (n=277)	Gp1 PH+PAWP>15 mm Hg (n=48)	Cpc PH HFpEF (n=75)	*P* value
Echocardiography				
LV end-diastolic volume, mL (n=273)	80.7±24.8	82.9±36.8	94.1±29.1†	0.01
LV mass, g (n=367)	131.8±44.0*	160.5±54.3	169.9±56.0	<0.0001
LV mass index, g/m² (n=376)	69.8±20.0*	79.2±22.7	80.2±21.1	0.0002
LA volume, mL (n=363)	40.7±17.2*	52.4±25.3*	65.5±26.3*	<0.0001
LA volume index, mL/m² (n=333)	22.8 [18.6–27.6]	25.7 [19.7–31.9]	29.8 [23.7–40.1]*	<0.0001
LV ejection fraction, % (n=374)	61.7±7.3	59.7±11.9	57.0±8.5†	0.0001
LV global longitudinal strain, % (n=239)	18.4±3.0*	15.6±4.1	16.9±2.3	<0.0001
Lateral e′, cm/s (n=359)	10.4±3.6	10.2±4.0	8.2±2.8*	0.0001
Lateral E/e′ (n=345)	6.5 [5.3–9.1]	8.1 [5.9–10.6]	13.3 [9.4–18.8]*	<0.0001
RV end-diastolic area, cm²(n=332)	28.2±8.9	33.3±10.2*	26.2±7.2	0.0003
RV PA systolic pressure, mm Hg (n=339)	66.9±23.3	71.7±22.8	57.1±18.6*	0.001
RV FAC, % (n=331)	30.3±10.2	28.5±10.9	33.0±8.9	0.07
RV FAC/PASP, %/mm (n=303)	0.54±0.35	0.43±0.24‡	0.65±0.30	0.0008
RV free wall strain, % (n=286)	18.7±6.3	17.1±5.2	20.0±5.7	0.13
RV free wall strain/PASP, %/mm (n=260)	0.28 [0.18–0.41]	0.23 [0.17–0.33]‡	0.35 [0.23–0.49]	0.03

Values represent mean (SD) or median (25th, 75th). *P* values were computed using overall group ANOVA or Kruskal-Wallis test with post hoc comparisons using Tukey or Steel-Dwass, respectively. Cpc indicates combined pre- and postcapillary; FAC, fractional area change; Gp1, group 1; HFpEF, heart failure with preserved ejection fraction; LA, left atrial; LV, left ventricular; PA, pulmonary artery; PASP, pulmonary artery systolic pressure; PAWP, pulmonary artery wedge pressure; PH, pulmonary hypertension, PVDOMICS, redefining pulmonary hypertension through pulmonary vascular disease phenomics; and RV, right ventricular.

**P*<0.05 vs all.

†*P*<0.05 vs Gp1 PH+PCWP≤15 mm Hg.

‡*P*<0.05 vs Cpc PH HFpEF.

Functional Status and Resting Hemodynamics in PAH and HFpEF

Patients with Cpc PH HFpEF had the greatest symptom burden (higher New York Heart Association class, worse quality of life) and poorest exercise tolerance (peak oxygen consumption and 6-minute walk distance) (Table S1). Functional impairment was largely comparable in the 2 group 1 PH subgroups, regardless of resting PAWP.

By definition, group 1 PH+high PAWP patients had higher PAWP compared with group 1 PH+normal PAWP, which was comparable with those with Cpc PH HFpEF. In contrast to the mean PAWP, there was a progressive increase in the amplitude of the PAWP V wave across groups, with the Cpc PH HFpEF group having the greatest V wave, indicating poorest LA compliance (Figure 1; Table 4). Elevation of PVR was greatest in both group 1 PH subgroups compared to Cpc PH HFpEF. As a consequence of the combined elevation in resting PAWP and PVR, patients with group 1 PH+high PAWP had the highest resting mPA pressure compared with other groups (Table 4). There was a positive correlation between PAWP and mPA pressure in group 1 PH overall (r, +0.39; *P*<0.0001), with directionally similar correlation when restricted to group 1 PH+high PAWP (r, +0.27; *P*=0.06). Estimated LV stiffness was lowest in the group 1 PH+normal PAWP group versus comparable results in group 1 PH+high PAWP and Cpc PH HFpEF (Figure S2).

Response to Hemodynamic Provocation

Following 100% O_2 and inhaled nitric oxide, there was a greater reduction in PAWP in group 1 PH+high PAWP compared with Cpc PH HFpEF and group 1 PH+normal PAWP, but there remained a progressive gradient of PAWP abnormalities across the 3 groups (Table 4; Table S4; Figure 1) Other group differences in hemodynamics, including larger PAWP V waves in Cpc PH HFpEF, persisted from baseline through the O_2, nitric oxide, and fluid challenge phases (Table 4; Table S4; Figure 1).

Considering repeated measurements across all phases, there were significant group differences in the change with provocation for PAWP, PAWP V wave, Δ V wave height, and mPA pressure. Baseline group differences in PVR persisted across repeated measurements (Figure 1). During exercise, PAWP increased progressively across the 3 groups, while exercise PVR was highest in the 2 group 1 PH groups (Table S4). Patients with Cpc PH HFpEF had the lowest peak VO_2 and most severe dyspnea with exercise, while the 2 group 1 PH

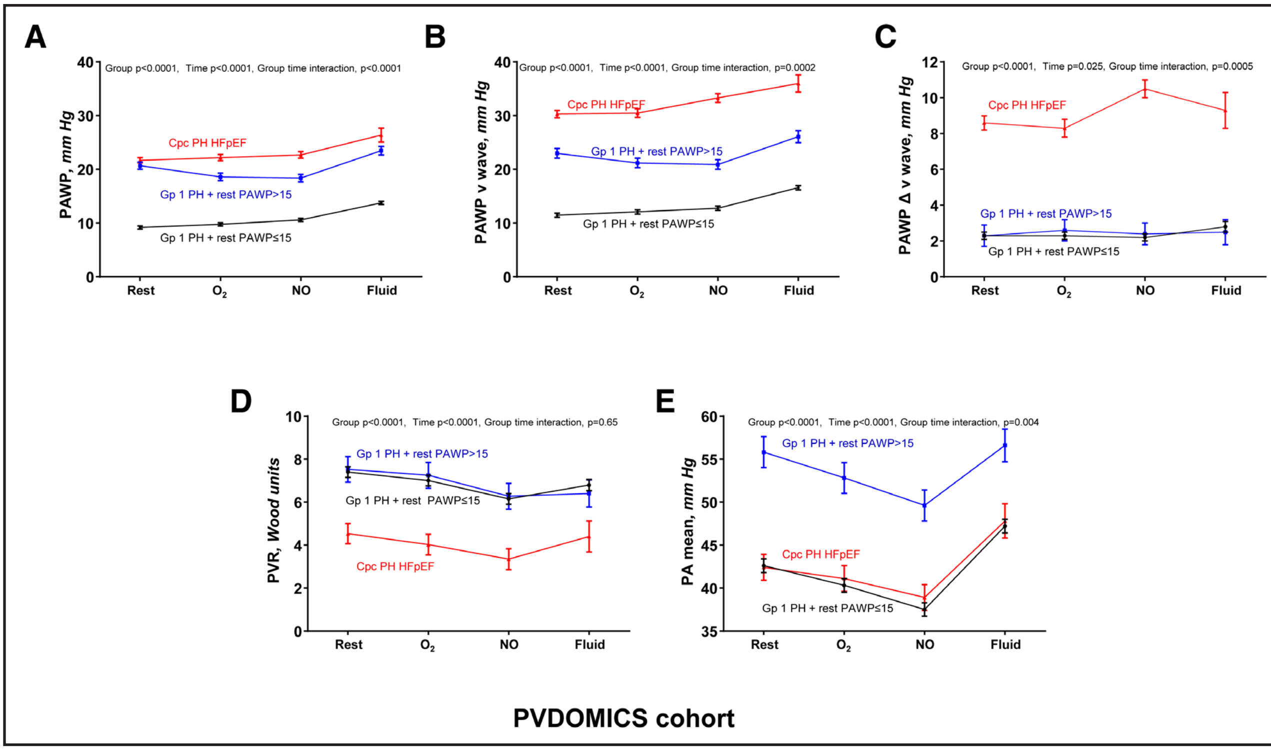

Figure 1. Dynamic hemodynamic provocation in group 1 pulmonary hypertension (PH) subgroups compared with combined pre- and postcapillary (Cpc) pulmonary hypertension (PH) heart failure with preserved ejection fraction (HFpEF).

Combined pre- and postcapillary PH heart failure with preserved ejection fraction was associated with the highest overall pulmonary artery wedge pressure (PAWP) averaged across all phases of provocation, including oxygen, nitric oxide, and fluid challenge (**A**), along with the highest PAWP V wave (**B**) and greatest rise in V wave during atrial filling (**C**) compared with both group 1 PH subgroups. Patients with group 1 PH regardless of resting PAWP had the highest PVR compared with combined pre- and postcapillary PH heart failure with preserved ejection fraction (**D**). Due to the combination of high PAWP and severe pulmonary vascular resistance (PVR) elevation, patients with group 1 PH+high PAWP had the highest mean PA pressure across all phases (**E**). Values represent mean and SE computed using mixed models. Gp1 indicates group 1; and PA, pulmonary artery.

subgroups had the greatest ventilatory inefficiency (Table S4).

Metabolome Differences

Metabolomic analyses revealed little difference between group 1 PH with high versus normal PAWP, with only 24 metabolites differing after false discovery rate correction, with modest effect size, and none remaining significant after adjusting for BMI (Figure 2; Supplemental Material). In contrast, there were large differences in the metabolome between group 1 PH+high PAWP versus Cpc PH HFpEF, with 100 metabolites different after false discovery rate correction, of which 91 remained significant after BMI adjustment (Supplemental Material).

Compared with group 1 PH+high PAWP, Cpc PH HFpEF was associated with greater deficiency of amino acids associated with metabolic dysfunction (including glycine and serine), nitric oxide signaling (arginine) and cellular senescence (tryptophan), among others (see the full list of altered metabolites in the Supplemental Material). Pathway enrichment analysis showed that, compared with group 1 PH+high PAWP, downregulated metabolites in the Cpc PH HFpEF group were mainly involved in amino acid metabolism–related pathways (Figure S4). Notably, 15 of the 20 proteogenic amino acids (arginine, asparagine, aspartate, cysteine, glutamine, glycine, histidine, isoleucine, lysine, methionine, phenylalanine, proline, serine, threonine, and tryptophan) were decreased in the Cpc PH HFpEF group (Supplemental Material). There were also elevated levels of amino acid catabolism–related metabolites, such as urea, *N*-acetylcitrulline, and homocitrulline, suggesting increased liver degradation through the urea cycle in Cpc PH HFpEF, contributing to amino acid deficiency distinct from group 1 PH+high PAWP.[28]

Outcomes

Over a median follow-up of 4.7 years (interquartile range, 4.0–6.0), there were 107 deaths in the group 1 PH and Cpc PH HFpEF cohorts. Patients with Cpc PH HFpEF had worse unadjusted survival (log-rank P=0.01) (Figure S5). However, after adjusting for baseline differences in age, NYHA functional class, 6-minute walk distance, and NT-proBNP (N-terminal pro-B-type natriuretic peptide), there were no significant group differences in survival

Table 4. Rest Hemodynamics in Group 1 Pulmonary Hypertension by Resting Pulmonary Artery Wedge Pressure and Combined Pre- and Postcapillary Pulmonary Hypertension Heart Failure With Preserved Ejection Fraction in the PVDOMICS Cohort

	Gp1 PH+PAWP≤15 mm Hg (n=277)	Gp1 PH+PAWP>15 mm Hg (n=48)	Cpc PH HFpEF (n=75)	*P* value
Rest hemodynamics				
Systolic blood pressure, mm Hg (n=395)	120.9±22.2	125.6±23.8	142.6±22.8*	<0.0001
RA mean, mm Hg (n=400)	6.4±4.0*	12.8±4.8	13.5±4.7	<0.0001
PAWP mean, mm Hg (n=400)	9.2±3.3*	20.7±4.4	21.7±4.5	<0.0001
PAWP V wave, mm Hg (n=397)	11.5±3.9*	23.1±5.2*	30.3±9.3*	<0.0001
PAWP V − mean, mm Hg (n=397)	2.3±2.0	2.3±1.9	8.6±7.6*	<0.0001
V_{30} index, mL/m² (n=248)	80.6 [65.6–99.2]*	69.0 [59.2–74.7]	69.8 [57.8–85.7]	0.0004
Beta stiffness (n=248)	5.84 [5.79–5.89]*	6.31 [6.09–6.77]	6.37 [3.04–6.77]	<0.0001
Left atrial compliance	10.5 [5.8–17.7]	9.7 [6.7–15.4]	4.3 [3.1–7.3]*	<0.0001
PA, mean, mm Hg (n=400)	42.6±13.7	55.8±13.5*	42.4±9.5	<0.0001
Cardiac output, L/min (n=395)	5.2±1.7	5.4±1.9	5.0±1.5	0.35
Cardiac index, L/min/m² (n=395)	2.8±0.8	2.7±1.0	2.4±0.7†	0.0008
PVR, Wood units (n=395)	6.3 [4.1–9.4]	6.3 [4.5–10.1]	3.7 [2.6–5.6]*	<0.0001
Change in hemodynamics with nitric oxide				
Δ RA mean, mm Hg (n=324)	−0.4±2.1	−1.9±3.3*	−0.3±2.8	0.002
Δ PAWP mean, mm Hg (n=358)	+1.5±4.0	−2.3±5.6*	+1.4±4.4	<0.0001
Δ PAWP V wave, mm Hg (n=354)	+1.4±4.2*	−2.0±6.1*	+3.7±7.0*	<0.0001
Δ PAWP V, mean, mm Hg (n=354)	−0.1±2.7	+0.2±2.7	+2.3±5.8*	<0.0001
Δ PA mean, mm Hg (n=363)	−5.1±4.9	−6.0±5.9	−3.7±6.6	0.09
Δ PVR, Wood units (n=349)	−0.8 [−2.1, −0.7]	−0.7 [−2.1, −0.0]	−1.1 [−1.8, −0.3]	0.68
Δ Cardiac output, L/min (n=353)	−0.2 [−0.7, +0.2]*	−0.1 [−0.5, +0.9]	−0.1 [−0.4, +0.4]	0.001

Values represent mean (SD) or median (25th, 75th). *P* values were computed using overall group ANOVA or Kruskal-Wallis test with post hoc comparisons using Tukey or Steel-Dwass, respectively. Cpc indicates combined pre- and postcapillary; Gp1, group 1; HFpEF, heart failure with preserved ejection fraction; PA, pulmonary artery; PAWP, pulmonary artery wedge pressure; PH, pulmonary hypertension, PVDOMICS, redefining pulmonary hypertension through pulmonary vascular disease phenomics; PVR, pulmonary vascular resistance; RA, right atrial; V_{30}, estimated left ventricular end-diastolic volume at common pulmonary artery wedge pressure of 30 mm Hg.

**P*<0.05 vs all.

†*P*<0.05 vs Gp1 PH + PAWP≤15 mm Hg.

(Cpc PH HFpEF versus group 1 PH+normal PAWP hazard ration [HR], 0.56 [95% CI, 0.30–1.05]; *P*=0.07; Cpc PH HFpEF vs group 1 PH+high PAWP HR, 0.49 [95% CI, 0.21–1.14]; *P*=0.10]

Validation Cohort 1: Group 1 PH and Cpc PH HFpEF With Exercise RHC (2006–2024)

Validation cohort 1 included 402 patients with PH (31% [n=124] group 1 PH, 69% [n=278] Cpc PH). Of the 124 patients with group 1 PH, 18% (n=22) had blinded end-expiratory PAWP at rest >15 mm Hg, with comparably severe PVR elevation across both group 1 PH subgroups. Most group 1 patients represented an incident new diagnosis of PH in this cohort (79% [n=81] with group 1 PH+normal PAWP versus 68% [n=15] with group 1 PH+high PAWP). Patients with Cpc PH HFpEF had greater age, BMI, hypertension, and atrial fibrillation burden compared with group 1 PH, and those with group 1 PH+normal PAWP had the most normal left heart remodeling/function by echocardiography (Table S5).

Like the PVDOMICS cohort, exercise PAWP and V wave abnormalities were of intermediate severity in group 1 PH+high PAWP compared with the other 2 groups (group *P*<0.0001, interaction *P*<0.0001) (Table 5; Figure 3; Figure S6).Patients with group 1 PH+high PAWP had a greater rise in RA/PAWP ratio and decrease in LVTMP without prominent PAWP V waves compared with Cpc PH HFpEF, supporting a contribution from enhanced pericardial interaction to the observed exercise PAWP elevation (Figure 3). However, the elevation in exercise PAWP in group 1 PH+high PAWP was not merely related to pericardial restraint alone, as exercise PAWP remained normal in group 1 PH+normal PAWP despite similar pericardial interaction during exercise with comparably low LVTMP and high RA/PAWP, supporting an additional contribution from left heart diastolic dysfunction to the PAWP abnormalities in group 1 PH+high PAWP (Figure 3).

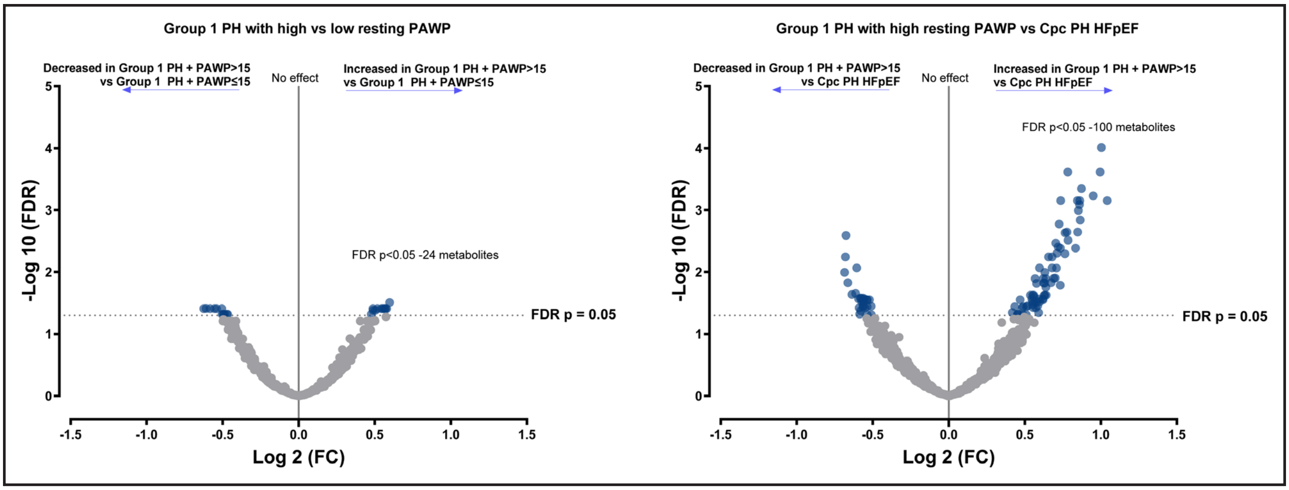

Figure 2. Metabolome in group 1 pulmonary hypertension (PH) with or without high pulmonary artery wedge pressure (PAWP) compared with group 1 PH with or without normal PAWP and combined pre- and postcapillary (Cpc) PH heart failure with preserved ejection fraction (HFpEF).

There was little difference in the metabolome of group 1 PH+high PAWP compared with group 1 PH+normal PAWP. In contrast, the metabolome of group 1 PH+high PAWP was markedly different from that seen in patients with overt combined pre- and postcapillary PH heart failure with preserved ejection fraction, supporting important biological differences between the 2 groups. Blue dots represent differences that remain significant, with false discovery rate (FDR)–corrected $P<0.05$. FC indicates fold change.

Validation Cohort 2: Group 1 PH With Exercise RHC (2024–2025)

Validation cohort 2 included 55 patients with adjudicated group 1 PH of whom 22% (n=12) had high PAWP at rest. Group 1 PH+high PAWP remained associated with greater obesity, with comparable prevalence of previously diagnosed group 1 PH with prior use of vasodilators (51% [n=22] with normal PAWP versus 41% [n=5] with high PAWP, $P=0.56$) (Table S6). Similar to other cohorts, PAWP during exercise was higher in group 1 PH+high PAWP ($P<0.0008$) compared with group 1 PH+normal PAWP without a prominent rise in PAWP V waves ($P=0.73$) (Table S7).

Validation Cohort 3: Resting RHC With Prior or Current PAH Therapy Use (2000–2020)

In the validation cohort 3 (n=787), there were 665 adjudicated cases of group 1 PH, of whom 19% (n=126) had a high PAWP at rest, and 122 adjudicated cases of group 2 PH, of whom 18% (n=22) had a normal PAWP at rest. Group 1 PH+high PAWP had greater obesity compared with group 1 PH+normal PAWP, with patients with Cpc PH HFpEF having older age, greater atrial fibrillation, and more abnormal left heart remodeling by echocardiography. Consistent with other cohorts, there was a progressive gradient of PAWP abnormalities observed across the 3 groups, with highest pulmonary artery (PA) pressure in group 1 PH+high PAWP (Table S8).

Sensitivity and Exploratory Analyses

In the PVDOMICS cohort, adjustment for baseline comorbidity differences (age, hypertension, diabetes, sleep apnea, and BMI) and PH treatment history (number of years of PH before enrollment and prior PH medication use) still resulted in persistent differences in PAWP and LV remodeling in group 1 PH+high PAWP, suggesting that comorbidity differences and prior PH therapy alone did not explain group differences in left heart hemodynamics (Table S9). Consistent with this, the prevalence of high PAWP in group 1 PH was comparable among prevalent and incident cases in validation cohorts 1 and 2 (21.8% [n=12] versus 17.7% [n=22], $P=0.52$) while being slightly more common among prevalent compared with incident cases in validation cohort 3 (26.6% [n=57] versus 15.3% [n=69], $P=0.0006$). Among patients who underwent repeat RHC in validation cohorts 1 and 2 (n=65), there was greater reduction with PH therapy in PAWP, RA pressure, and PA pressure in group 1 PH+high PAWP compared with group 1 PH+normal PAWP, with comparable improvement in PVR, cardiac output, and NT-proBNP in both groups. These differences in longitudinal hemodynamic response were replicated in those from validation cohort 3 with repeat RHC assessment (n=328) (Table S10).

Diagnostic Agreement

Among the Mayo Clinic validation cohort 1 (n=402), there was good agreement between 2 independent adjudicators (Y.N.V.R. and W.R.M.) of the diagnosis of group 1 PH or Cpc PH HFpEF when considering all available information, including hemodynamics, in a probabilistic manner (kappa coefficient, 0.89 [95% CI, 0.84–0.94]; $P<0.0001$). Among the subset of PVDOMICS participants from the Mayo Clinic where independent readjudication was possible (n=54) (R.P.F. and Y.N.V.R.), there

Table 5. Rest, Feet Up, and Exercise Hemodynamics in Validation Cohort 1 With Micromanometer End-Expiratory Pressure Measurement

	Gp1 PH+PAWP≤15 mm Hg (n=102)	Gp1 PH+PAWP>15 mm Hg (n=22)	Cpc PH HFpEF (n=278)	*P* value
Rest				
RA, mean, mm Hg (n=402)	8.9±3.6*	13.2±4.0	14.0±4.4	<0.0001
PAWP, mean, mm Hg (n=402)	11.2±2.8*	17.9±1.6*	21.8±4.0*	<0.0001
PAWP V wave, mm Hg (n=402)	11.9±3.4*	20.4±3.5*	34.3±10.8*	<0.0001
PAWP V − mean, mm Hg (n=402)	0.0 [0.0–1.0]*	1.0 [0.0–5.3]	12.0 [5.0–18.0]*	<0.0001
RA/PAWP (n=402)	0.70 [0.60–1.00]	0.74 [0.62–0.86]	0.63 [0.52–0.75]*	<0.0001
LVTMP, mm Hg (n=402)	2.3±3.9*	4.7±4.0*	7.8±4.3*	<0.0001
PA mean, mm Hg (n=402)	39.5±13.3	49.4±10.9*	38.0±9.2	<0.0001
PVR, Wood units (n=402)	5.2 [3.3–7.5]	7.0 [4.1–8.2]	3.1 [2.4–4.5]*	<0.0001
Cardiac output, L/min (n=402)	5.1±1.8	5.1±2.0	4.4±1.2†	0.0001
Feet Up				
RA mean, mm Hg (n=321)	11.8±4.4*	16.5±6.0	18.2±5.5	<0.0001
PAWP mean, mm Hg (n=354)	14.8±4.3*	21.1±4.1*	26.2±4.6*	<0.0001
PAWP V wave, mm Hg (n=354)	15.9±5.4*	24.2±6.4*	41.5±12.0*	<0.0001
PAWP V − mean, mm Hg (n=354)	0.0 [0.0–2.0]*	1.0 [0.0–5.5]*	15.0 [7.0–22.5]*	<0.0001
RA/PAWP (n=321)	0.70 [0.57–1.00]	0.72 [0.59–0.93]	0.69 [0.56–0.82]	0.23
LVTMP, mm Hg (n=321)	3.1±5.4	4.8±5.9	7.9±5.4*	<0.0001
Peak exercise				
RA, mean, mm Hg (n=368)	17.1±6.0*	22.6±6.7	25.6±7.3	<0.0001
PAWP, mean, mm Hg (n=402)	18.9±4.6*	23.8±4.9*	33.6±6.3*	<0.0001
PAWP V wave, mm Hg (n=402)	20.1±5.2*	26.8±5.6*	50.0±12.9*	<0.0001
PAWP V − mean, mm Hg (n=402)	1.0 [0.0–2.0]	2.0 [0.0–5.0]	16.0 [7.0–25.0]*	<0.0001
RA/PCWP (n=368)	0.88 [0.70–1.08]	0.98 [0.78–1.06]	0.73 [0.60–0.91]*	<0.0001
LVTMP, mm Hg (n=368)	1.8±6.6	1.2±4.8	8.2±7.7*	<0.0001
PA mean, mm Hg (n=402)	56.6±15.0*	67.0±17.3*	52.7±10.4*	<0.0001
PVR, Wood units (n=384)	4.0 [2.5–6.8]	5.9 [3.8–8.1]	2.7 [1.7–3.9]*	<0.0001
Cardiac output, L/min (n=384)	8.4±2.8	8.1±3.2	6.7±2.3*	<0.0001

Values represent mean (SD) or median (25th, 75th). P values were computed using overall group ANOVA or Kruskal-Wallis test with post hoc comparisons using Tukey or Steel-Dwass, respectively. Cpc indicates combined pre- and postcapillary; Gp1, group 1; HFpEF, heart failure with preserved ejection fraction; LVTMP, left ventricular transmural pressure; PAWP, pulmonary artery wedge pressure; PH, pulmonary hypertension, PVDOMICS, redefining pulmonary hypertension through pulmonary vascular disease phenomics; PVR, pulmonary vascular resistance; and RA, right atrial.

**P*<0.05 vs all.

†*P*<0.05 vs Gp1 PH+PAWP≤15 mm Hg.

was also good agreement in the diagnosis of group 1 PH or Cpc PH HFpEF (kappa coefficient, 0.92 [95% CI, 0.82–1.00]; *P*<0.0001). Among participants initially adjudicated to have group 1 PH, the percent agreement for diagnosis of group 1 PH between 2 independent reviewers was 93.9%, 86.3%, 96.4%, and 84.5% in adjudicated patients from the primary cohort and validation cohorts 1 to 3, respectively.

DISCUSSION

In this study, we report the first comprehensive phenotypic characterization of prospectively adjudicated patients with group 1 PH stratified by resting PAWP. We identify that, when the diagnosis of group 1 PH is established by PH experts, around 1 of 5 patients has a true PAWP elevation at rest that would normally be indicative of group 2 PH. Elevation in PAWP in this group persisted across repeated provocative measurements, and findings were replicated in 3 independent validation cohorts. Although the identified patients with group 1 PH+high PAWP would fulfill current diagnostic criteria for Cpc PH due to resting elevation in PAWP and PVR,[3] their clinical characteristics, prior pulmonary vasodilator use, and metabolomic profile were more comparable with traditional group 1 PH with normal PAWP. However, there were also key differences, including greater obesity, higher pretest probability for HFpEF, subclinical abnormalities in LV remodeling, and diastolic dysfunction by multiple independent measures. The observed

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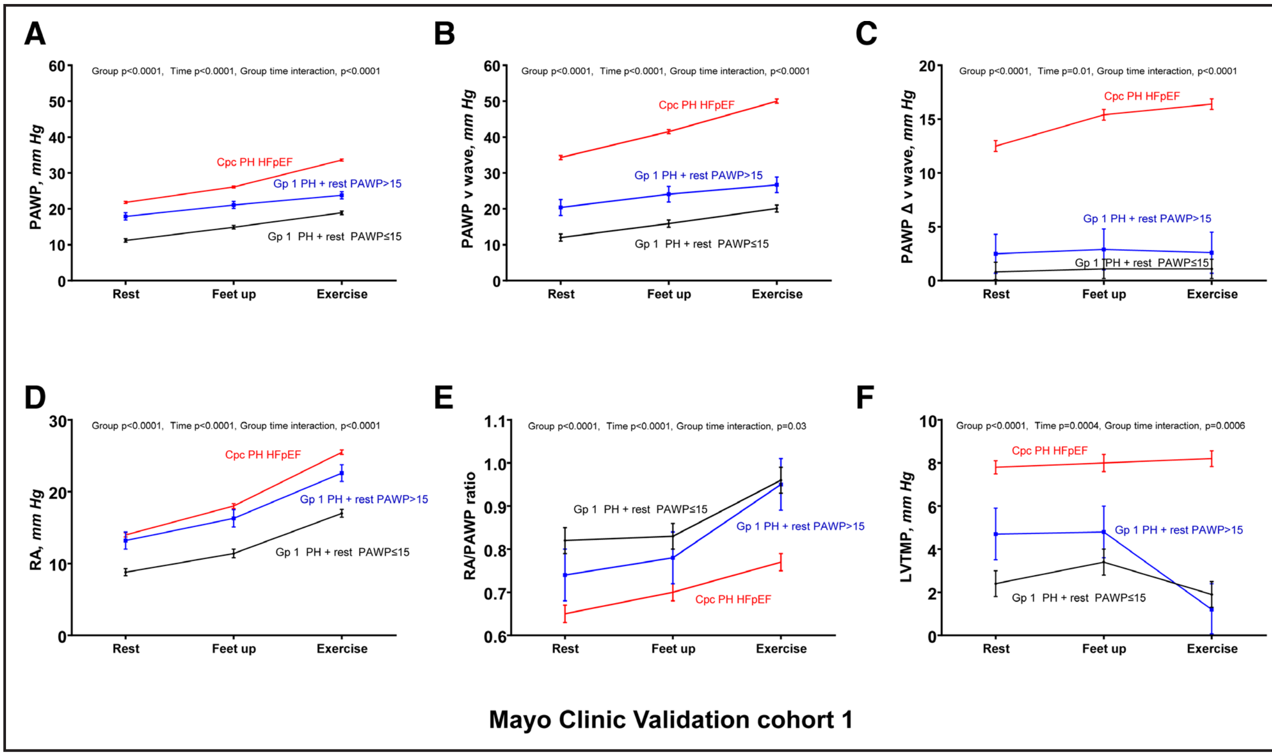

Figure 3. Dynamic hemodynamic provocation with feet up and exercise challenge in group 1 pulmonary hypertension (PH) subgroups compared with combined pre- and postcapillary (Cpc) PH heart failure with preserved ejection fraction (HFpEF) in Mayo Clinic validation cohort 1.

Combined pre- and postcapillary PH heart failure with preserved ejection fraction was associated with the highest overall pulmonary artery wedge pressure (PAWP) across rest, feet up, and exercise phases, along with the highest PAWP V wave (**A** through **C**) compared with both group 1 PH subgroups. Patients with group 1 PH+high PAWP had elevated right atrial (RA) pressure with provocation with evidence of enhanced pericardial interaction contributing to the observed exercise PAWP elevation, with a rise in right atrial/PAWP ratio and decrease in left ventricular transmural pressure (LVTMP) without prominent V waves compared with combined pre- and postcapillary PH heart failure with preserved ejection fraction (**D** through **F**). However, the elevation in exercise PAWP was not merely related to diastolic ventricular interaction alone, as patients with group 1 PH+normal PAWP at rest had similar exercise right atrial/PAWP and left ventricular transmural pressure changes suggestive of ventricular interaction, but absolute PAWP remained lowest in this group. Collectively, these data support the likely contribution of both diastolic dysfunction and ventricular interaction to the observed exercise PAWP elevation in group 1 PH+high PAWP. Values represent mean and SE computed using mixed models. Gp 1 indicates group 1.

left heart changes in group 1 PH+high PAWP were pathologic relative to healthy controls but of intermediate severity between true Cpc PH HFpEF and more traditionally defined group 1 PH.

Despite some similarities in LV abnormalities between Cpc PH HFpEF and group 1 PH+high PAWP, there were also notable differences, including less severe pulmonary vascular disease, worse LA compliance, and higher PAWP V wave response during atrial filling in Cpc PH. The metabolome differed between Cpc PH HFpEF and group 1 PH+high PAWP, with greater evidence of amino acid deficiency in Cpc PH HFpEF, further supporting key biological differences. Exercise PAWP response in group 1 PH+high PAWP was intermediately abnormal relative to the other 2 groups, with evidence of enhanced ventricular interaction additionally contributing to abnormal PAWP elevation during exercise. Collectively, these data support the idea that group 1 PH+high PAWP is pathophysiologically distinct from Cpc PH HFpEF, and although these patients appear more closely aligned with a diagnosis of group 1 PH, the unique pathophysiology inherent to also having PAWP elevation suggests that they are also distinct from traditionally defined group 1 PH, calling for further study in trials focused on this subgroup of patients.

Group 1 PH With High Resting PAWP: A Unique Subphenotype of Group 1 PH

The present data support the hypothesis that patients with group 1 PH+high PAWP truly have precapillary pulmonary vascular pathology. While there is superimposed LV diastolic dysfunction, this is likely not the primary basis for the observed severe PVR elevation. Although recent guidelines and position statements have endorsed the need for probabilistic evaluation of the diagnosis of group 1 PH versus Cpc PH HFpEF, without excessive reliance on resting PAWP thresholds alone to dichotomize the diagnosis,[4,5] the identified group 1 PH subset with PAWP>15 mm Hg has been systematically excluded

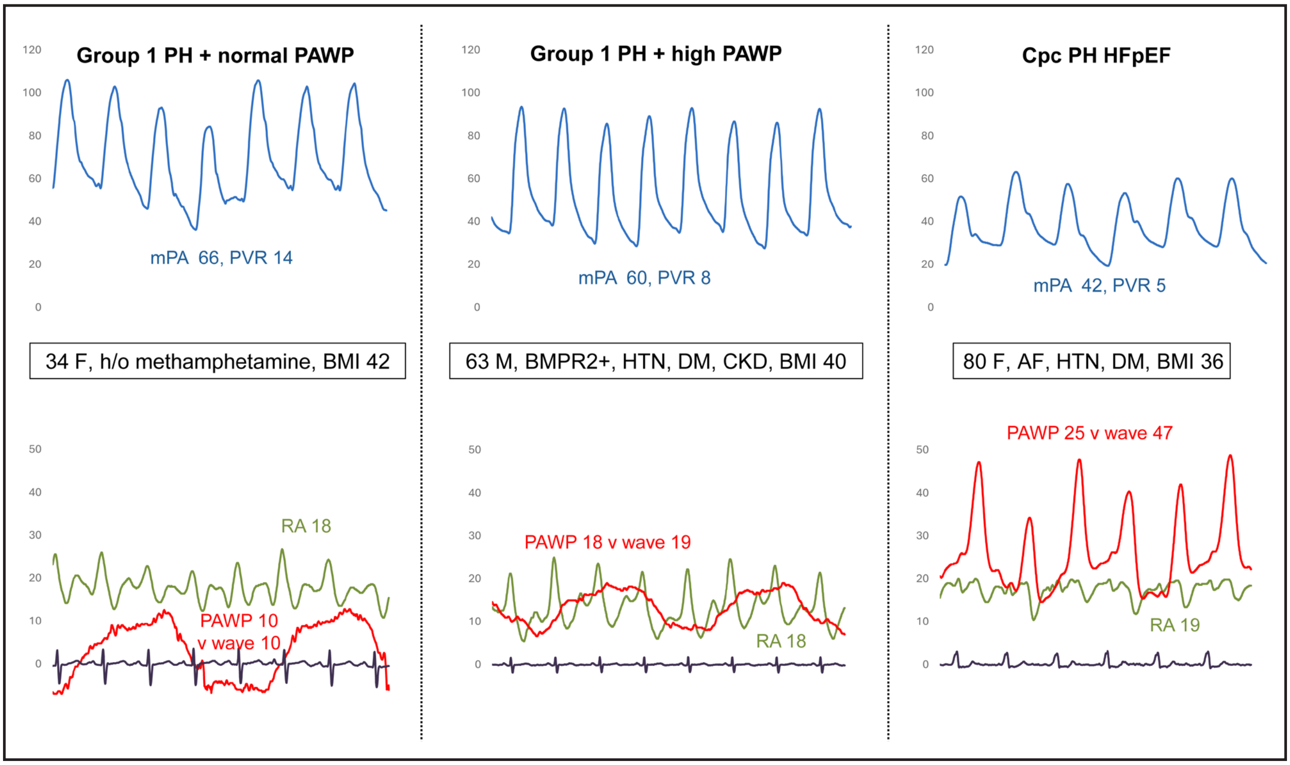

Figure 4. Representative tracings with group 1 pulmonary hypertension (PH) with normal pulmonary artery wedge pressure (PAWP), group 1 PH with high PAWP, and combined pre- and postcapillary (Cpc) PH heart failure with preserved ejection fraction (HFpEF).

The participant in the **middle** with group 1 PH+high PAWP had severe PH with a "flat" elevation in PAWP that equalized with right atrial (RA) pressure along with prominent inspiratory decline consistent with a component of enhanced pericardial restraint contributing to the PAWP elevation. Participants with combined pre- and postcapillary PH heart failure with preserved ejection fraction (**right**), on average, demonstrated larger PAWP V waves compared with group 1 PH with less severe PH. Patients with group 1 PH+normal PAWP had the lowest PAWP with severe PH. AF indicates atrial fibrillation; BMI, body mass index; BMPR2+, bone morphogenetic protein receptor type 2; CKD, chronic kidney disease; DM, diabetes mellitus; F, female; h/o, history of; HTN, hypertension; M, male; mPA, mean pulmonary artery; and PVR, pulmonary vascular resistance.

from clinical trials, so there is inadequate evidence to guide therapy in this subset.

Post hoc analyses of randomized trials have suggested no interaction based on baseline PAWP,[10] although the systematic exclusion of those with resting PAWP>15 mm Hg limits the generalizability of these findings. Observational studies have suggested a worse therapeutic response in those with adjudicated group 1 PH who develop a higher resting PAWP after vasodilator therapy initiation.[29] The observation of high baseline vasodilator use among patients with group 1 PH+high PAWP in this study might support the hypothesis that chronic pulmonary vasodilator use and improved LV filling may have unmasked latent LV diastolic dysfunction in some patients with group 1 PH. Although this may contribute in some cases, the generally comparable prevalence of PAWP elevation even in incident group 1 PH argues against this being the only explanation for the observed PAWP elevation. Although the mechanisms of PAWP elevation require further study, it seems plausible that a subset of these patients with prevalent longstanding group 1 PH may develop independent acquired left heart dysfunction from HFpEF with associated risk factors of obesity and aging. It is also possible that there may be still unidentified common risk factors that may contribute to both PAH and HFpEF and predispose to disproportionate pulmonary vascular remodeling with even modest left heart pathology in a subset of patients.

Although likely contributory, the presence of HFpEF risk factors alone appears insufficient to explain the observed PAWP elevation. One additional contributor appears to be enhanced diastolic ventricular interaction from the right heart impacting LV filling, as suggested by the flat PAWP waveforms without prominent V waves, evidence of pericardial restraint during exercise, and PH therapy–induced reduction in both RA and PAWP in group 1 PH+high PAWP. There may also be pulmonary venous remodeling that could contribute to modest overestimation of LA pressure when using PAWP.[30] Indeed, the observed acute and chronic reduction in PAWP with pulmonary vasodilators in group 1 PH+high PAWP might support this hypothesis as a reflection of pulmonary venodilation.[31] Collectively, it seems likely that multiple independent pathophysiological mechanisms may variably contribute to the modest PAWP elevation in some patients with group 1 PH (Figure 4).

Therapeutic Potential for HFpEF Therapies

Additionally, given that the group 1 PH+high PAWP group had the worst overall PA pressure elevation due to the superimposed impact of an increased PAWP on group 1 PH–related PVR elevation, and the correlations observed between PAWP and PA pressure even in group 1 PH, there is a need for further study of therapeutic approaches to lower PAWP in these patients. Some patients with group 1 PH may develop LV cardiomyocyte atrophy from chronic LV underfilling,[32] which could contribute to elevated risk of LV chamber noncompliance that may manifest following pulmonary vasodilator–induced improved LV filling or acquisition of superimposed HFpEF risk factors.[33–35] There was a notably high burden of excess adiposity in group 1 PH+high PAWP, and with sodium glucose cotransporter 2 inhibitors and incretins demonstrating clear adipose–reducing effects with associated heart failure benefits in patients with overt HFpEF,[36–39] further study of the added benefit of these therapies on top of pulmonary vasodilators in group 1 PH+high PAWP is needed.

With sodium glucose cotransporter 2 inhibitors, there is clear evidence of adipose tissue regression[36,37,40] with associated resting and dynamic PAWP reduction,[36,41–44] with improved dynamic RV-PA coupling.[17,42] The effect of incretin therapy on PAWP remains unknown, but weight loss has been associated with improved PAWP and PH,[45] and clinical trials of incretins in overt HFpEF have demonstrated improvement in natriuretic peptides, myocardial remodeling, and functional status.[38,39,46–48] Therefore, although there is increasing interest and guideline emphasis in identifying occult PAWP abnormalities with provocation among group 1 PH with normal resting PAWP,[5,8] the current data suggest that a clinically important subset of patients with group 1 PH has overt elevation of PAWP even at rest, emphasizing the need for integrated probabilistic evaluation of all available information, including hemodynamics, imaging, comorbidities and potentially even metabolomics,[49] to establish a clinical diagnosis of PAH. As a result of being excluded from all prior trials of group 1 PH, there is a pressing need for randomized trials to determine the optimal treatment strategies in this unique group 1 PH subgroup with high PAWP at rest.

Strengths and Limitations

The present analysis has multiple strengths, including multiple validation cohorts, multicenter representation, manual readjudication of hemodynamic tracings, comprehensive functional and phenotypic characterization, metabolomics assessment, and diagnostic adjudication of all cases by a PH specialist. The prevalence of PAWP elevation in group 1 PH is higher than previously reported,[50] which may relate to greater prevalence of obesity, and manual readjudication of PAWP measurement in all cases, with findings replicated with 4 independent cohorts. Although our approach to classify patients as having group 1 PH or Cpc PH HFpEF was consistent with current recommendations to use all available information in a probabilistic manner to make a binary decision regarding the most likely diagnosis, this process is inherently subjective, and there is currently no existing operational framework to guide application. We therefore considered a series of clinical and hemodynamic criteria (Table 1; see details in the Supplemental Material) in each case through detailed chart/medical history review with manual hemodynamic tracing adjudication by a PH physician, and made a probabilistic diagnosis in each individual case of the presence of either group 1 PH or Cpc PH HFpEF. Although this approach is subjective by definition, it is consistent with contemporary clinical practice enhancing generalizability. We also tested concordance statistics between independent PH experts and demonstrated good diagnostic agreement for our adjudication process. However, it must be acknowledged that there is currently no objective "gold-standard" to fully separate group 1 PH in the presence of risk factors for HFpEF from Cpc HFpEF, and we cannot exclude the possibility that some of the patients with group 1 PH+high PAWP may truly have a diagnosis of Cpc PH, and in many ways this is a key finding of this study. Future efforts are therefore needed to develop and test more objective multivariable-based clinical criteria to separate group 1 PH from Cpc PH HFpEF without reliance on current binary hemodynamic thresholds alone. Acknowledging these limitations, the finding of large differences between group 1 PH+high PAWP and Cpc PH (and large similarities between group 1 PH with high or normal PAWP), using unbiased metabolomics, adjudication agreement by independent PH experts, and validation of our results in 3 independent cohorts, supports the likelihood that most of the study patients with group 1 PH+high PAWP had true group 1 PH as the primary pathology.

Conclusion

Although a resting PAWP>15 mm Hg is commonly used to implicate left heart failure as the cause of PH, about 1 in 5 patients with adjudicated group 1 PH has elevated resting PAWP, which appears to be related in part to modest left heart diastolic dysfunction that is less severe than overt Cpc PH HFpEF. These patients display metabolomic signatures more typical of group 1 PH than Cpc PH, suggesting that they should be classified as precapillary predominant PH, but the pathophysiological differences present underline the priority for dedicated trials to guide treatment moving forward.

ARTICLE INFORMATION

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Author Contributions
Dr Reddy, conceptualization, methodology, formal analysis, investigation, data curation, writing – original draft, writing – review and editing, visualization, and supervision; Dr Miranda, data curation and writing – review and editing; Dr Varma, data curation and writing – review and editing; Dr Asokan, formal analysis and writing – review and editing; Dr Borlaug, conceptualization, methodology, and writing – review and editing; all authors, data curation and writing – review and editing.

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REFERENCES

1. Zeder K, Avian A, Mak S, Giannakoulas G, Kawut SM, Maron BA, Humbert M, Olschewski H, Kovacs G. Pulmonary arterial wedge pressure in healthy subjects: a meta-analysis. *Eur Respir J*. 2024;64:2400967. doi: 10.1183/13993003.00967-2024
2. Humbert M, Kovacs G, Hoeper MM, Badagliacca R, Berger RMF, Brida M, Carlsen J, Coats AJS, Escribano-Subias P, Ferrari P, et al; ESC/ERS Scientific Document Group. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. *Eur Heart J*. 2022;43:3618–3731. doi: 10.1093/eurheartj/ehac237
3. Kovacs G, Bartolome S, Denton CP, Gatzoulis MA, Gu S, Khanna D, Badesch D, Montani D. Definition, classification and diagnosis of pulmonary hypertension. *Eur Respir J*. 2024;64:2401324. doi: 10.1183/13993003.01324-2024
4. Rayner SG, Tedford RJ, Leary PJ, Mak S, Houston BA. "This Patient Needs a Doctor, Not a Guideline!" The zone of uncertainty in pulmonary arterial wedge pressure measurement. *Am J Respir Crit Care Med*. 2024;210:712–714. doi: 10.1164/rccm.202402-0359VP
5. Maron BA, Bortman G, De Marco T, Huston JH, Lang IM, Rosenkranz SH, Vachiéry J-L, Tedford RJ. Pulmonary hypertension associated with left heart disease. *Eur Respir J*. 2024;64:2401344. doi: 10.1183/13993003.01344-2024
6. Hoeper MM. Navigating uncertainty in PAH classification: wedge pressure versus disease phenotypes. *Circulation*. 2025;152:4–6. doi: 10.1161/CIRCULATIONAHA.125.073885
7. Frost AE, Farber HW, Barst RJ, Miller DP, Elliott CG, McGoon MD. Demographics and outcomes of patients diagnosed with pulmonary hypertension with pulmonary capillary wedge pressures 16 to 18 mm Hg: insights from the REVEAL registry. *Chest*. 2013;143:185–195. doi: 10.1378/chest.11-1387
8. Reddy YNV, Frantz RP, Hassoun PM, Hemnes AR, Horn E, Leopold JA, Rischard F, Rosenzweig EB, Hill NS, Erzurum SC, et al. Clinical implications of pretest probability of HFpEF on outcomes in precapillary pulmonary hypertension. *J Am Coll Cardiol*. 2024;84:2196–2210. doi: 10.1016/j.jacc.2024.08.061
9. LeVarge BL, Pomerantsev E, Channick RN. Reliance on end-expiratory wedge pressure leads to misclassification of pulmonary hypertension. *Eur Respir J*. 2014;44:425–434. doi: 10.1183/09031936.00209313
10. Kovacs G, Moutchia J, Zeder K, Maron BA, Al-Naamani N, Ventetuolo C, Olschewski H, Kawut SM. Clinical response to pulmonary arterial hypertension treatment does not depend on pulmonary arterial wedge pressure: a meta-analysis using individual participant data from randomized clinical trials. *Am J Respir Crit Care Med*. 2024;210:844–847. doi: 10.1164/rccm.202403-0612RL
11. Hemnes AR, Leopold JA, Radeva MK, Beck GJ, Abidov A, Aldred MA, Barnard J, Rosenzweig EB, Borlaug BA, Chung WK, et al; PVDOMICS Study Group. Clinical characteristics and transplant-free survival across the spectrum of pulmonary vascular disease. *J Am Coll Cardiol*. 2022;80:697–718. doi: 10.1016/j.jacc.2022.05.038
12. Reddy YNV, Carter RE, Sundaram V, Kaye DM, Handoko ML, Tedford RJ, Andersen MJ, Sharma K, Obokata M, Verbrugge FH, et al. An evidence-based screening tool for heart failure with preserved ejection fraction: the HFpEF-ABA score. *Nat Med*. 2024;30:2258–2264. doi: 10.1038/s41591-024-03140-1
13. Tang WHW, Wilcox JD, Jacob MS, Rosenzweig EB, Borlaug BA, Frantz RP, Hassoun PM, Hemnes AR, Hill NS, Horn EM, et al. Comprehensive diagnostic evaluation of cardiovascular physiology in patients with pulmonary vascular disease: insights from the PVDOMICS program. *Circ Heart Fail*. 2020;13:e006363. doi: 10.1161/CIRCHEARTFAILURE.119.006363
14. Reddy YNV, Obokata M, Verbrugge FH, Lin G, Borlaug BA. Atrial dysfunction in patients with heart failure with preserved ejection fraction and atrial fibrillation. *J Am Coll Cardiol*. 2020;76:1051–1064. doi: 10.1016/j.jacc.2020.07.009
15. Reddy YNV, El-Sabbagh A, Nishimura RA. Comparing pulmonary arterial wedge pressure and left ventricular end diastolic pressure for assessment of left-sided filling pressures. *JAMA Cardiol*. 2018;3:453–454. doi: 10.1001/jamacardio.2018.0318
16. Hemnes AR, Beck GJ, Newman JH, Abidov A, Aldred MA, Barnard J, Berman Rosenzweig E, Borlaug BA, Chung WK, Comhair SAA, et al; PVDOMICS Study Group. PVDOMICS: A multi-center study to improve understanding of pulmonary vascular disease through phenomics. *Circ Res*. 2017;121:1136–1139. doi: 10.1161/CIRCRESAHA.117.311737
17. Reddy YNV, Carter RE, Sorimachi H, Omar M, Popovic D, Alogna A, Jensen MD, Borlaug BA. Dapagliflozin and right ventricular-pulmonary vascular interaction in heart failure with preserved ejection fraction: a secondary analysis of a randomized clinical trial. *JAMA Cardiol*. 2024;9:843–851. doi: 10.1001/jamacardio.2024.1914
18. Reddy YNV, Carter RE, Obokata M, Redfield MM, Borlaug BA. A simple, evidence-based approach to help guide diagnosis of heart failure with preserved ejection fraction. *Circulation*. 2018;138:861–870. doi: 10.1161/CIRCULATIONAHA.118.034646

19. Reddy YNV, Obokata M, Koepp KE, Egbe AC, Wiley B, Borlaug BA. The β-adrenergic agonist albuterol improves pulmonary vascular reserve in heart failure with preserved ejection fraction. *Circ Res*. 2019;124:306–314. doi: 10.1161/CIRCRESAHA.118.313832
20. Obokata M, Reddy YNV, Pislaru SV, Melenovsky V, Borlaug BA. Evidence supporting the existence of a distinct obese phenotype of heart failure with preserved ejection fraction. *Circulation*. 2017;136:6–19. doi: 10.1161/CIRCULATIONAHA.116.026807
21. Borlaug BA, Reddy YNV. The Role of the pericardium in heart failure: implications for pathophysiology and treatment. *JACC Heart Fail*. 2019;7:574–585. doi: 10.1016/j.jchf.2019.03.021
22. Omote K, Sorimachi H, Obokata M, Reddy YNV, Verbrugge FH, Omar M, DuBrock HM, Redfield MM, Borlaug BA. Pulmonary vascular disease in pulmonary hypertension due to left heart disease: pathophysiologic implications. *Eur Heart J*. 2022;43:3417–3431. doi: 10.1093/eurheartj/ehac184
23. Cubero Salazar IM, Lancaster AC, Jani VP, Montovano MJ, Kauffman M, Weller A, Ambale-Venkatesh B, Zimmerman SL, Simpson CE, Kolb TM, et al. Poor cardiac output reserve in pulmonary arterial hypertension is associated with right ventricular stiffness and impaired interventricular dependence. *Eur Respir J*. 2024;64:2400420. doi: 10.1183/13993003.00420-2024
24. Zhou M, Shao J, Wu C-Y, Shu L, Dong W, Liu Y, Chen M, Wynn RM, Wang J, Wang J, et al. Targeting BCAA catabolism to treat obesity-associated insulin resistance. *Diabetes*. 2019;68:1730–1746. doi: 10.2337/db18-0927
25. Newgard CB, An J, Bain JR, Muehlbauer MJ, Stevens RD, Lien LF, Haqq AM, Shah SH, Arlotto M, Slentz CA, et al. A branched-chain amino acid-related metabolic signature that differentiates obese and lean humans and contributes to insulin resistance. *Cell Metab*. 2009;9:311–326. doi: 10.1016/j.cmet.2009.02.002
26. Ho JE, Larson MG, Ghorbani A, Cheng S, Chen M-H, Keyes M, Rhee EP, Clish CB, Vasan RS, Gerszten RE, et al. Metabolomic profiles of body mass index in the framingham heart study reveal distinct cardiometabolic phenotypes. *PLoS One*. 2016;11:e0148361. doi: 10.1371/journal.pone.0148361
27. Xia J, Wishart DS. MSEA: a web-based tool to identify biologically meaningful patterns in quantitative metabolomic data. *Nucleic Acids Res*. 2010;38:W71–W77. doi: 10.1093/nar/gkq329
28. Chandel NS. Amino acid metabolism. *Cold Spring Harb Perspect Biol*. 2021;13:a040584. doi: 10.1101/cshperspect.a040584
29. Harder EM, Divo MJ, Washko GR, Leopold JA, Rahaghi FN, Waxman AB. Implications of mean pulmonary arterial wedge pressure trajectories in pulmonary arterial hypertension. *Am J Respir Crit Care Med*. 2024;209:316–324. doi: 10.1164/rccm.202306-1072OC
30. Tada A, Omote K, Fayyaz AU, Simard TJ, Miranda WR, Naser JA, Reddy YNV, Redfield MM, Borlaug BA. Pulmonary arterial wedge pressure exceeding left atrial pressure: clinical implications. *Eur Heart J*. 2025;46:4123–4126. doi: 10.1093/eurheartj/ehaf392
31. Rivera-Estrada C, Saltzman PW, Singer D, Katz LN. Action of hypoxia on the pulmonary vasculature. *Circ Res*. 1958;6:10–14. doi: 10.1161/01.res.6.1.10
32. Manders E, Bogaard HJ, Handoko ML, van de Veerdonk MC, Keogh A, Westerhof N, Stienen GJ, Dos Remedios CG, Humbert M, Dorfmuller P, et al. Contractile dysfunction of left ventricular cardiomyocytes in patients with pulmonary arterial hypertension. *J Am Coll Cardiol*. 2014;64:28–37. doi: 10.1016/j.jacc.2014.04.031
33. Kosmala W, Jedrzejuk D, Derzhko R, Przewlocka-Kosmala M, Mysiak A, Bednarek-Tupikowska G. Left ventricular function impairment in patients with normal-weight obesity: contribution of abdominal fat deposition, profibrotic state, reduced insulin sensitivity, and proinflammatory activation. *Circ Cardiovasc Imaging*. 2012;5:349–356. doi: 10.1161/CIRCIMAGING.111.969956
34. Ferraro S, Perrone-Filardi P, Desiderio A, Betocchi S, D'Alto M, Liguori L, Trimigliozzi P, Turco S, Chiariello M. Left ventricular systolic and diastolic function in severe obesity: a radionuclide study. *Cardiology*. 1996;87:347–353. doi: 10.1159/000177118
35. Selvaraj S, Martinez EE, Aguilar FG, Kim K-YA, Peng J, Sha J, Irvin MR, Lewis CE, Hunt SC, Arnett DK, et al. Association of central adiposity with adverse cardiac mechanics: findings from the Hypertension Genetic Epidemiology Network Study. *Circ Cardiovasc Imaging*. 2016;9:1. doi: 10.1161/circimaging.115.004396
36. Borlaug BA, Reddy YNV, Braun A, Sorimachi H, Omar M, Popovic D, Alogna A, Jensen MD, Carter R. Cardiac and metabolic effects of dapagliflozin in heart failure with preserved ejection fraction: the CAMEO-DAPA trial. *Circulation*. 2023;148:834–844. doi: 10.1161/CIRCULATIONAHA.123.065134
37. Naser JA, Tada A, Harada T, Reddy YNV, Carter RE, Testani JM, Jensen MD, Borlaug BA. Effects of dapagliflozin on body composition and its relation to hemodynamics in heart failure with preserved ejection fraction. *Circulation*. 2024;149:2026–2028. doi: 10.1161/CIRCULATIONAHA.124.069479
38. Borlaug BA, Kitzman DW, Davies MJ, Rasmussen S, Barros E, Butler J, Einfeldt MN, Hovingh GK, Møller DV, Petrie MC, et al. Semaglutide in HFpEF across obesity class and by body weight reduction: a prespecified analysis of the STEP-HFpEF trial. *Nat Med*. 2023;29:2358–2365. doi: 10.1038/s41591-023-02526-x
39. Packer M, Zile MR, Kramer CM, Baum SJ, Litwin SE, Menon V, Ge J, Weerakkody GJ, Ou Y, Bunck MC, et al; SUMMIT Trial Study Group. Tirzepatide for heart failure with preserved ejection fraction and obesity. *N Engl J Med*. 2025;392:427–437. doi: 10.1056/NEJMoa2410027
40. Reddy YNV, Melenovsky V, Asokan AK, Haluzik M, Carter RE, Nair S, Jensen MD, Borlaug BA. Dapagliflozin, peptide YY, and weight loss in heart failure with preserved ejection fraction. *Eur Heart J*. 2024;45:3889–3891. doi: 10.1093/eurheartj/ehae534
41. Nassif ME, Windsor SL, Borlaug BA, Kitzman DW, Shah SJ, Tang F, Khariton Y, Malik AO, Khumri T, Umpierrez G, et al. The SGLT2 inhibitor dapagliflozin in heart failure with preserved ejection fraction: a multicenter randomized trial. *Nat Med*. 2021;27:1954–1960. doi: 10.1038/s41591-021-01536-x
42. Nassif ME, Qintar M, Windsor SL, Jermyn R, Shavelle DM, Tang F, Lamba S, Bhatt K, Brush J, Civitello A, et al. Empagliflozin effects on pulmonary artery pressure in patients with heart failure: results from the EMBRACE-HF trial. *Circulation*. 2021;143:1673–1686. doi: 10.1161/CIRCULATIONAHA.120.052503
43. Tada A, Burkhoff D, Naser JA, Harada T, Pourmussa B, Reddy YNV, Jensen MD, Carter RE, Demmer RT, Testani JM, et al. Dapagliflozin enhances arterial and venous compliance during exercise in heart failure with preserved ejection fraction: insights from the CAMEO-DAPA trial. *Circulation*. 2024;150:997–1009. doi: 10.1161/CIRCULATIONAHA.124.068788
44. Doi S, Reddy YNV, Jensen MD, Smith JR, Borlaug BA. Dapagliflozin and ventilatory control during exercise in heart failure with preserved ejection fraction: the CAMEO-DAPA trial. *Eur Heart J*. 2025;46:1554–1556. doi: 10.1093/eurheartj/ehaf027
45. Reddy YNV, Anantha-Narayanan M, Obokata M, Koepp KE, Erwin P, Carter RE, Borlaug BA. Hemodynamic effects of weight loss in obesity: a systematic review and meta-analysis. *JACC Heart Fail*. 2019;7:678–687. doi: 10.1016/j.jchf.2019.04.019
46. Kramer CM, Borlaug BA, Zile MR, Ruff D, DiMaria JM, Menon V, Ou Y, Zarante AM, Hurt KC, Murakami M, et al. Tirzepatide reduces LV mass and paracardiac adipose tissue in obesity-related heart failure: SUMMIT CMR substudy. *J Am Coll Cardiol*. 2025;85:699–706. doi: 10.1016/j.jacc.2024.11.001
47. Solomon SD, Ostrominski JW, Wang X, Shah SJ, Borlaug BA, Butler J, Davies MJ, Kitzman DW, Verma S, Abildstrøm SZ, et al; STEP-HFpEF Trial Committees and Investigators. Effect of semaglutide on cardiac structure and function in patients with obesity-related heart failure. *J Am Coll Cardiol*. 2024;84:1587–1602. doi: 10.1016/j.jacc.2024.08.021
48. Borlaug BA, Zile MR, Kramer CM, Baum SJ, Hurt K, Litwin SE, Murakami M, Ou Y, Upadhyay N, Packer M. Effects of tirzepatide on circulatory overload and end-organ damage in heart failure with preserved ejection fraction and obesity: a secondary analysis of the SUMMIT trial. *Nat Med*. 2025;31:544–551. doi: 10.1038/s41591-024-03374-z
49. Reddy YNV, Asokan AK, Frantz RP, Hemnes A, Hassoun PM, Barnard J, Horn E, Leopold JA, Rischard F, Rosenzweig EB, et al. Metabolomic evidence of biological overlap with heart failure with preserved ejection fraction in a subset of pulmonary arterial hypertension. *Am J Respir Crit Care Med*. 2025;211:1701–1713. doi: 10.1164/rccm.202501-0034OC
50. Park D-H, Fuge J, Kamp JC, Harrigfeld B, Berliner D, Hoeper MM, Olsson KM. Reassessing pulmonary hypertension classification: utilizing criteria for heart failure with preserved ejection fraction instead of pulmonary arterial wedge pressure. *J Clin Med*. 2024;13:7582. doi: 10.3390/jcm13247582
51. Obokata M, Reddy YNV, Pislaru SV, Melenovsky V, Borlaug BA. Evidence supporting the existence of a distinct obese phenotype of heart failure with preserved ejection fraction. *Circulation*. 2017;136:6–19. doi: 10.1161/CIRCULATIONAHA.116.026807